Supplementary Information

Title: Auditing PPI Graph Augmentation for Missense Variant Pathogenicity Prediction under Topology, Connectivity and Homology Constraints

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Notes on scope

This Supplementary Information provides reconstruction details, split statistics, model comparisons, uncertainty estimates, topology-stratified summaries and negative-control results. Full figure source data, topology-stratified metrics, corrected-sample statistics and fusion-control summaries are included in the prepared compact source-data archive.

Figure guide: Main Figure 1B–F uses matched representation experiments; Tables 11–12 supply the performance summaries and ensemble intervals, and Table 4A supplies both resampling intervals for Figure 1C. The partial-ablation curve in Figure 1E is detailed in Table 11B. Main Figure 2B/D/E uses matched fusion or representation results from Table 11, and Figure 2C uses the paired ensemble intervals in Table 12. Figure 3A shows ESM similarity, while Figure 3B–F uses matched original-split ensembles for the topology analysis and protein-clustered intervals in Table 5; Figure 4 uses matched original-split ensembles for the transition and feature profiles in Tables 6 and 10. Tables 2–3, 4B–C and 8B preserve separate reference analyses and their protocols. Matched configurations and reporting conventions are specified in Table 8A and Methods 5. External-source and BioGRID validation are documented in Methods 6, Tables 14–17 and main Figure 5. Different fitted prediction sets should not be interchanged.

Supplementary Methods

Supplementary Methods 1. Dataset reconstruction and source files

We reconstructed a missense variant pathogenicity dataset from the ClinVar `variant_summary.txt.gz` table obtained from the official NCBI ClinVar FTP `tab_delimited` directory. The present analysis uses retained, checksum-identified processed tables and model inputs; their exact historical ClinVar content-release date is unverified. Checksums and verified row identities identify the frozen evaluated inputs but do not establish temporal independence. We retained germline GRCh38 single-nucleotide variant records and extracted protein-level HGVS annotations. Missense substitutions were identified using a strict three-letter amino-acid substitution pattern.

Pathogenic, Likely pathogenic and Pathogenic/Likely pathogenic records were grouped as pathogenic. Benign, Likely benign and Benign/Likely benign records were grouped as benign. Supported review status was defined as the exact ClinVar `ReviewStatus` strings `criteria provided, single submitter`, `criteria provided, multiple submitters, no conflicts`, `reviewed by expert panel`, and `practice guideline`; records outside this whitelist were excluded. Conflicting VariationIDs were removed, and duplicate VariationIDs were collapsed after conflict removal. The final mapped dataset contained 177,213 filtered, non-conflicting ClinVar missense variants with criteria-supported review status, including 127,953 benign and 49,260 pathogenic variants.

ClinVar gene symbols were mapped to STRING human protein nodes using upper-cased gene symbols and the STRING protein alias table. Only genes that mapped uniquely to a STRING protein present in the STRING links graph were retained. Mapping was performed at the protein-identifier level; isoform-specific effects and transcript-level ambiguity were not modeled separately. The reconstructed ClinVar-STRING dataset covered 14,123 unique variant-labelled STRING proteins embedded in a larger STRING node universe containing 19,622 proteins. The final variant tables retained genomic coordinates, reference and alternate VCF alleles, VariationID, binary label, gene symbol and mapped STRING protein identifier.

CADD and AlphaMissense scores were matched to ClinVar variants by GRCh38 genomic coordinates and reference/alternate alleles. CADD GRCh38 v1.7 whole-genome SNV scores were matched for 177,205 of 177,213 variants, leaving eight missing CADD rows. AlphaMissense hg38 scores were matched for 171,753 variants, leaving 5,460 missing AlphaMissense rows. CADD-only and AlphaMissense-only baselines used their corresponding matched scalar scores. CADD+AlphaMissense models used CADD PHRED, AlphaMissense pathogenicity and an AlphaMissense-missing indicator. Raw-ESM fusion concatenated these scalar features with the 1,280-dimensional raw ESM protein representation, whereas graph-fusion models concatenated the same scalar features with the 1,280-dimensional FPP representation. Follow-up graph-control fusions used the same scalar CADD+AlphaMissense features concatenated with real FPP, direct train-test edge-ablated FPP, or topology-permuted FPP representations.

Supplementary Methods 2. Protein representations and graph propagation

Protein node features were generated with the Facebook ESM-2 t33 650M UR50D checkpoint. Residue-level embeddings were mean-pooled to obtain one 1,280-dimensional vector per protein. Proteins longer than 1,022 amino acids were split into chunks and combined by length-weighted averaging. The final feature table covered all 19,622 STRING graph proteins, with no missing graph proteins.

The primary representation-propagation graph was built from STRING v12 human protein links and contained 19,622 nodes and 13,715,404 directed edge-index entries in the retained model input object before adding self-loops. FPP representations were generated using fixed two-hop GCN-style propagation rather than an end-to-end trainable graph neural network. This design intentionally isolated topology-induced smoothing from capacity gains due to trainable graph layers, so the propagated features were treated as a controlled representation rather than an optimized end-to-end GNN predictor. The propagation adjacency retained edge-count multiplicities after adding self-loops, symmetrizing the retained edge list and summing duplicate contributions; STRING combined scores were not used as propagation weights. For the historical full FPP and all current matched graph arms, self-loops are added before symmetrization: A=(E+I)+(E+I)^T, retaining summed edge-count multiplicities after coalescing. Here E is the edge-index count matrix, D is the weighted degree matrix and X is the raw ESM feature matrix. Old ablation/permutation controls instead added a single loop after reverse edges. Their outputs remain historical; new matched controls correct this convention mismatch. We used symmetric normalization,

$$\hat{A} = D^{-1/2} A D^{-1/2}$$

and formed the propagated representation as

$$H = (X + \hat{A}X + \hat{A}^2X) / 3$$

FPP denotes fixed protein-feature propagation, the fixed two-hop operation called PPI-GCN in the main text. Both terms refer to the same non-trainable propagation, not to a trained graph neural network. This operation produced one propagated representation per STRING node and did not use variant labels during propagation. The full graph includes held-out features and nodes: this is transductive feature access, not use of held-out labels. All embedding-only variant inputs are identical within each mapped protein.

Two supplementary fixed operators were used. Fixed mean propagation applies row-normalized two-hop mean propagation with residual averaging. Fixed cosine-weighted propagation applies two rounds of cosine-softmax-weighted neighbour aggregation with residual averaging, including self-neighbours and softmax weights from the current feature cosine similarity. Both run under no-gradient computation and have no learned aggregation or attention parameters. These are fixed operator controls, not trained GraphSAGE/GAT baselines; trained graph models were not evaluated.

For topology profiling, we used a separate thresholded STRING graph based on combined score. The primary topology analyses used a combined-score threshold of 400, yielding 19,622 nodes and 929,472 graph edges. Degree, k-core, community membership, betweenness, clustering coefficient and community-boundary role were computed on this thresholded graph. The representation-propagation graph and the topology-description graph were related STRING-derived objects but not identical analysis objects.

For topology-permutation controls, the graph edges were relabelled by a random node permutation (`edge_index_perm = perm[edge_index]`), while ESM features, labels and train/validation/test variant assignments remained tied to the original protein identities. Variant features were then extracted from the propagated representation of each variant's original mapped STRING protein after propagation on the

permuted graph. The graph edge count and unlabeled topology were preserved, but the original protein-to-neighbourhood correspondence was broken. Three graph permutations were combined with three downstream classifier seeds, and the downstream MLP was retrained for each permutation/seed pair.

Supplementary Methods 3. Evaluation splits, homology-aware stress test and statistical analysis

The original evaluation split was constructed at the STRING protein level using seed 42 and an approximate 8:1:1 train, validation and test ratio. Variants assigned to the same STRING protein were kept in the same partition. This produced 141,864 training variants, 17,129 validation variants and 18,220 test variants, with no shared proteins across partitions. The corresponding numbers of unique proteins were 11,298, 1,412 and 1,413 in the train, validation and test sets. All proteins remained embedded in the same full STRING association graph, so train and test proteins could still be connected through STRING edges.

For homology-aware evaluation, STRING v12 human protein sequences were clustered with MMseqs using minimum sequence identity 0.40, coverage threshold 0.50 and cov-mode 0. Homologous clusters rather than individual proteins were assigned to train, validation and test partitions, with no protein or homology-cluster overlap across partitions. This split was used as a stricter stress test for reduced sequence-related sharing across partitions, not as a prevalence-matched replacement for the original protein-level split. It preserved the original row counts: 141,864 training variants, 17,129 validation variants and 18,220 test variants. The homology-aware split contained 7,370, 3,315 and 3,438 unique proteins in the train, validation and test sets, respectively, and was built from 9,731 protein clusters with mean cluster size 1.45 and maximum cluster size 251. Because test prevalence changed from 31.95% in the original split to 10.56% in the homology-aware split, AUPRC values should be interpreted within each split rather than directly compared across splits. Protein-cluster assignment sorted clusters by descending variant-row count and greedily chose the partition with largest remaining target capacity (tie order: train, validation, test), without class-prevalence stratification. The implemented commands are:

```
mmseqs createdb all_split_proteins.fa split_proteins_db
mmseqs cluster split_proteins_db split_proteins_clu tmp \
  --min-seq-id 0.40 -c 0.50 --cov-mode 0 --threads 8
mmseqs createtsv split_proteins_db split_proteins_db \
  split_proteins_clu split_proteins_cluster.tsv
```

The retained script does not pin the MMseqs binary version or non-overridden defaults; exact regeneration remains dependent on that historical environment. Cluster-disjointness under these parameters is not a guarantee that every distant homolog is separated. Historical single-run fusion did not establish repeated-run inferiority; the new three-seed comparison is in Table 11; class prevalence affects cross-split comparison, not comparisons on the identical within-split test set.

Variant-level pathogenicity was predicted with validation-selected multilayer perceptron classifiers. For representation-based and fusion models, each variant inherited the representation of its mapped STRING protein, and the classifier used two hidden layers with 512 and 128 units, ReLU activations, dropout 0.3 and a single-logit output. Low-dimensional external-score-only baselines used a smaller 32/16 classifier with dropout 0.2, while architecture-matched CADD+AlphaMissense and fusion-control comparisons used the same 512/128 hidden-layer architecture and training protocol. Input-layer parameter counts differ between score-only and fusion models because their input dimensions differ. The nominal main-model protocol was BCEWithLogitsLoss and positive-class weighting, using Adam with learning rate 1e-3, weight decay 1e-5, batch size 1024, a maximum of 50 epochs and early stopping after five consecutive validation checks without validation AUC improvement. The positive-class weight was computed from the training partition as `N_negative / N_positive`. The best epoch was selected by validation AUC.

In the evaluated MLP configurations, no additional z-score standardization was applied to CADD, AlphaMissense or embedding-derived inputs. Missing CADD and AlphaMissense values were imputed using training-set medians, and binary missingness indicators were retained as binary indicators. The same preprocessing and feature construction were used across raw ESM, edge-ablated FPP, topology-permuted FPP and real FPP fusion controls. The matched comparisons used seeds 42, 43 and 44 (Table 8A); topology-permuted fusion combined three sampled graph permutations with these classifier seeds. Archived reference protocols are specified in Table 8B. The relative fusion-control comparisons use the same hidden-layer architecture, training protocol and preprocessing, whereas absolute fusion performance may depend on feature-scaling choices that were not separately optimized.

Neighbour-risk, neighbour pathogenic-rate and neighbour training-variant-count features were computed from training-set labels only. Validation and test labels were not used to construct these topology-context features. When a test protein had multiple labelled training-neighbour variants, the neighbour pathogenic rate was variant-weighted rather than protein-weighted, so proteins with many labelled neighbouring variants contributed proportionally to the local training-label context.

Historical uncertainty was evaluated using 1,000 variant-level bootstrap resamples; current matched analyses use 2,000 (Methods 5). Historical analyses included variant resampling and protein-clustered bootstrap resampling. Variant-level bootstrap resampled individual test variants. Protein-clustered bootstrap resampled test proteins as clusters and retained all variants belonging to each sampled protein, thereby accounting for repeated variants from the same protein. Confidence intervals were reported as empirical 95% bootstrap intervals. Fusion bootstrap conditions on the retained fitted ensembles; it omits retraining and newly sampled graph-permutation uncertainty. An interval spanning zero does not establish equivalence or absence of a topology effect. Edge removal changes degrees and normalized propagation weights as well as direct paths; higher-order train-test paths and train-validation-test paths remain. Corrected-sample topology-feature means and effect sizes are reported descriptively because multiple variants from the same protein share network features and are not statistically independent (Table 6). No inferential significance is claimed for these profiles.

Supplementary Methods 4. Negative-control analysis

Exploratory controls included random raw features, random-feature graph propagation, label shuffling and feature-shuffled FPP propagation. Only the re-materialized feature-shuffled FPP control is formally reported here because it was retained with complete final-audit provenance. It was rerun using the same downstream evaluation protocol and achieved validation AUC 0.7080, validation AUPRC 0.4241, test AUC 0.5868 and test AUPRC 0.3707. The result is used only as supplementary negative-control evidence and is not part of the main comparative claims.

Supplementary Tables

Supplementary Table 1. Dataset, split and graph summary

Item	Train	Validation	Test	Total or value	Notes
Orig. variants	141,864	17,129	18,220	177,213	Protein split
Orig. benign	102,818	12,737	12,398	127,953	Label 0
Orig. pathogenic	39,046	4,392	5,822	49,260	Label 1
Orig. proteins	11,298	1,412	1,413	14,123 labelled proteins	No protein overlap
Homol. variants	141,864	17,129	18,220	177,213	Cluster split
Homol. benign	96,212	15,445	16,296	127,953	Label 0
Homol. pathogenic	45,652	1,684	1,924	49,260	Label 1
Homol. proteins	7,370	3,315	3,438	14,123 labelled proteins	No protein or cluster overlap
Homol. clusters	NA	NA	NA	9,731	Mean 1.45; max 251
Repr. graph	NA	NA	NA	19,622 nodes; 13,715,404 directed entries	Propagation graph
Topology graph	NA	NA	NA	19,622 nodes; 929,472 edges	STRING score >=400
CADD matching	NA	NA	NA	177,205 matched; 8 missing	GRCh38 v1.7 whole-genome SNV scores
AlphaMissense matching	NA	NA	NA	171,753 matched; 5,460 missing	hg38 AlphaMissense table

Orig. denotes the original protein-level split. Homol. denotes the homology-aware cluster split. Repr. denotes the representation-propagation graph.

Supplementary Table 2. Historical representation and architecture-control performance

Table 2 preserves separate raw-ESM and FPP reference estimates, historical repeated-run summaries and operator controls. The matched representation estimates in main Figure 1B/D/F and Figure 2D are supplied by Table 11A, and the partial-ablation series in Figure 1E by Table 11B. Their ensemble intervals are supplied by Table 12 and Table 4A. The reference rows in Table 2 should not be substituted for these matched comparisons.

Split	Model	Val AUC	Val AUPRC	Test AUC	Test AUPRC	Status
Orig.	Raw ESM	0.7814	0.5569	0.7790	0.6074	Reference
Orig.	FPP	0.8237	0.5821	0.7997	0.6496	Reference
Orig.	Raw ESM (3 seeds)	NA	NA	0.7835 +/- 0.0051	0.6116 +/- 0.0054	3 seeds
Orig.	FPP (3 seeds)	NA	NA	0.7950 +/- 0.0013	0.6484 +/- 0.0096	3 seeds
Orig.	Fixed mean propagation	0.8008	0.6136	0.7950	0.6044	Fixed operator
Orig.	Fixed cosine-weighted propagation	0.8005	0.5876	0.7949	0.6157	Fixed operator
Orig.	Degree-only	0.7298	0.4228	0.6377	0.3901	Control
Orig.	Feature-shuffled FPP	0.7080	0.4241	0.5868	0.3707	Rerun verified
Homol.	Raw ESM	0.7315	0.2171	0.7248	0.2418	Stress test
Homol.	FPP	0.7645	0.2450	0.7606	0.2768	Stress test

Orig. denotes the original split. Homol. denotes the homology-aware split. Arch. denotes architecture.

Supplementary Table 3. Historical external predictor and fusion-control performance

Table 3 preserves separate fusion-control estimates and single-tool reference context, including repeated classifier runs where available and a single-fit homology graph-fusion estimate. The matched fusion results in main Figure 2B/E are supplied by Table 11, and the paired ensemble intervals in Figure 2C by Table 12. The fitted prediction sets in Table 3 are not substituted for these matched comparisons.

Supplementary Table 3A. Original split

Model	n	Val AUC	Val AUPRC	Test AUC	Test AUPRC
CADD	1	0.9237	0.7855	0.9207	0.8049
AM	1	0.9597	0.9038	0.9470	0.8899
CADD+AM (arch.)	3	0.9667	0.9126	0.9572 +/- 0.00005	0.9068 +/- 0.0003
+ raw ESM	3	0.9686	0.9160	0.9680 +/- 0.0007	0.9317 +/- 0.0029
+ edge-ablated FPP	3	0.9728	0.9257	0.9708 +/- 0.0008	0.9395 +/- 0.0007
+ permuted FPP	9	0.9718	0.9232	0.9694 +/- 0.0008	0.9355 +/- 0.0015
+ real FPP	3	0.9732	0.9253	0.9718 +/- 0.0007	0.9415 +/- 0.0011

Notes for Table 3A: AM denotes AlphaMissense. "Arch." denotes the architecture-matched CADD+AlphaMissense baseline using the same 512/128 hidden-layer architecture and training protocol as the fusion controls; the input-layer parameter counts differ. The rows beginning with "+" add the listed protein representation to the CADD+AM feature set. Compared with the architecture-matched CADD+AM baseline, raw ESM fusion increased test AUC/AUPRC by +0.0108/+0.0249, edge-ablated FPP fusion by

+0.0136/+0.0327, topology-permuted FPP fusion by +0.0122/+0.0286 and real FPP fusion by +0.0146/+0.0347. The edge-ablated row uses one deterministic direct train-test edge-ablation graph evaluated across three classifier seeds, whereas the topology-permuted row combines three random graph permutations with three classifier seeds.

Supplementary Table 3B. Homology-aware split

Model	n	Val AUC	Val AUPRC	Test AUC	Test AUPRC
CADD	1	0.9506	0.6713	0.9504	0.6895
AM	1	0.9614	0.7945	0.9632	0.8029
CADD+AM (arch.)	3	0.9739	0.8362	0.9754 +/- 0.00005	0.8401 +/- 0.0002
+ raw ESM	3	0.9715	0.8228	0.9728 +/- 0.0003	0.8249 +/- 0.0016
+ FPP	1	0.9732	0.8273	0.9741	0.8333

The graph-fusion row in Table 3B is a separate single-fit stress-test result, not a fully symmetric repeated-ensemble comparison. Within Table 3B, its differences from the repeated-run architecture-matched CADD+AlphaMissense mean are -0.0013 AUC and -0.0068 AUPRC. Main Figure 2E instead uses three matched classifier fits per arm from Table 11 and the paired ensemble intervals in Table 12.

Supplementary Table 4. Uncertainty and repeated-run summary

Supplementary Table 4A. Matched raw ESM versus FPP uncertainty

Bootstrap	Metric	Delta	CI low	CI high	n
Variant	AUC	+0.0107	+0.0076	+0.0139	2,000
Variant	AUPRC	+0.0412	+0.0347	+0.0475	2,000
Protein-clustered	AUC	+0.0107	-0.0218	+0.0426	2,000
Protein-clustered	AUPRC	+0.0412	-0.0106	+0.0767	2,000

In Table 4A, Delta denotes the observed full-FPP-minus-raw-ESM difference on the same three-seed float64 ensemble predictions. Both resampling schemes use 2,000 paired replicates; the 95% intervals are the 2.5th and 97.5th percentiles of the corresponding bootstrap differences. These rows are the source for main Figure 1C and the protein rows agree with Table 12. The variant-resampling intervals do not account for within-protein dependence. The previous reference-pair bootstrap files remain unchanged in the dated archive and are not relabelled as matched results.

Supplementary Table 4B. Original-split strong-predictor fusion-control uncertainty

Comparison codes: C0, real FPP fusion versus architecture-matched CADD+AlphaMissense; C1, real FPP fusion versus raw ESM fusion; C2, real FPP fusion versus edge-ablated FPP fusion; C3, real FPP fusion versus topology-permuted FPP fusion.

ID	Metric	Delta	CI low	CI high	n
C0	AUC	+0.0143	+0.0062	+0.0224	1,000
C0	AUPRC	+0.0341	+0.0167	+0.0566	1,000
C1	AUC	+0.0038	+0.0005	+0.0075	1,000
C1	AUPRC	+0.0093	+0.0032	+0.0167	1,000
C2	AUC	+0.0009	-0.0009	+0.0027	1,000
C2	AUPRC	+0.0017	-0.0018	+0.0052	1,000
C3	AUC	+0.0019	-0.0005	+0.0052	1,000
C3	AUPRC	+0.0050	-0.0002	+0.0112	1,000

Supplementary Table 4C. Earlier retained-reference fusion bootstrap retained as context

Comparison code: R0, earlier retained-reference graph fusion versus earlier retained-reference CADD+AlphaMissense; not architecture-matched.

ID	Metric	Delta	CI low	CI high	n
R0	AUC	+0.0133	+0.0119	+0.0146	1,000
R0	AUPRC	+0.0310	+0.0268	+0.0350	1,000

Table 4A shows that variant-level resampling supports positive matched ensemble FPP gains, whereas protein-clustered resampling widens uncertainty. For its separate fitted prediction set, Table 4B shows that the real FPP fusion margin over raw ESM fusion remained positive under protein-clustered paired bootstrap, but the smaller margins over edge-ablated and topology-permuted graph controls crossed zero. Table 4C is retained as an earlier retained-reference context analysis and is not architecture-matched.

Table 4A supplies the matched ensemble intervals in main Figure 1C. Main Figure 2C uses the matched fusion intervals in Table 12; Tables 4B–C retain their separate fitted comparisons. Main Figure 2D and Figure 2E use three matched fits per arm for representations and fusion, respectively (Tables 11–12); the estimates in Tables 2 and 3B remain separate reference results.

Supplementary Table 5. Topology-stratified FPP gain over raw ESM

These results use matched raw-ESM and full-FPP ensembles, with probabilities averaged in float64 across classifier seeds 42–44. The original test memberships, topology buckets, 53 joint cells and four highlighted cells are fixed. The topology-description graph is distinct from the propagation graph (Methods 2). The same test set was used to examine the original cells and select the highlighted profiles; these profiles have not been independently confirmed in the external source.

Stratum	n	Positive rate	Delta AUC	Delta AUPRC
Degree-low	2,795	0.0426	+0.0185	+0.0050
Degree-mid	4,762	0.2089	+0.0135	+0.0195
Degree-high	10,663	0.4415	-0.0216	+0.0061
K-core-low	2,847	0.0425	+0.0175	+0.0044
K-core-mid	5,685	0.3078	+0.0603	+0.0960
K-core-high	9,688	0.4078	-0.0283	+0.0125
Risk-low	6,323	0.1056	+0.0231	+0.0474
Risk-mid	5,829	0.2882	+0.0334	+0.0617
Risk-high	6,019	0.5772	+0.0133	+0.0291
Betweenness-low	3,050	0.1393	+0.0545	+0.0781
Betweenness-mid	4,486	0.1899	+0.0368	+0.0417
Betweenness-high	10,684	0.4254	-0.0210	+0.0086
Clustering-low	8,234	0.2888	-0.0022	-0.0122
Clustering-mid	5,950	0.4198	+0.0343	+0.0471
Clustering-high	4,036	0.2344	+0.0309	+0.0902
Boundary-like	6,198	0.3527	+0.0026	+0.0076
Boundary-mixed	6,598	0.2963	+0.0060	+0.0587
Boundary-internal	5,397	0.3111	+0.0135	+0.0184

The risk strata contain 18,171 variants: 49 variants from 10 proteins had fewer than five labelled training-neighbour variants and were assigned the insufficient-signal code -1. The boundary strata contain 18,193 variants: 27 variants from five proteins had community ID -1 after communities with fewer than 50 nodes were excluded by the topology script. Of these, four variants mapped to two isolated proteins; the other 23 mapped to three non-isolated proteins. Their within-community fraction was undefined. Both omitted groups were smaller than the minimum group size of 50. Other strata cover all 18,220 test variants.

Selected exploratory joint topology cells:

Cell codes: J0, mid k-core / high betweenness; J1, mid k-core / mid clustering; J2, high neighbour risk / mid k-core; J3, high k-core / low clustering. The displayed cells are unchanged from the original selection.

ID	n	Pos. rate	Delta AUC	Delta AUPRC	AUC CI
J0	1,950	0.4297	+0.1115	+0.1202	-0.0012 to +0.1782
J1	2,089	0.4045	+0.0861	+0.0836	+0.0065 to +0.1445
J2	1,904	0.5767	+0.0832	+0.0686	-0.0177 to +0.1514
J3	4,791	0.3870	-0.0486	-0.0424	-0.2020 to +0.0312

J0–J3 contain 126, 163, 134 and 95 distinct proteins, respectively. Paired 95% intervals use 2,000 protein-clustered resamples within each fixed cell: sampled proteins contribute all their variants, with the same multiplicities for both models. Percentile intervals condition on the fitted ensembles and graph; they do not include retraining, graph-sampling or subgroup-selection uncertainty and are not adjusted for multiple comparisons. All 2,000 resamples were evaluable in the four displayed cells. The low-k-core/mid-betweenness, low-k-core/high-betweenness, low-k-core/high-clustering and mid-neighbour-risk/low-k-core cells had 2, 227, 95 and 724 single-class resamples, respectively; their undefined AUC/AP differences were excluded from the corresponding percentile intervals, without replacement draws. Full point estimates, protein counts, finite-replicate counts, intervals and memberships for all 53 cells, together with the 18 one-dimensional strata, are supplied in the compact source-data archive. These exploratory profiles do not independently confirm topology-defined benefit subgroups.

Supplementary Table 6. Descriptive corrected-sample and baseline-error-stratified profiles

Both comparisons use the matched original-split models: raw ESM versus full FPP, and architecture-matched CADD+AlphaMissense versus full-FPP fusion. Probabilities from classifier seeds 42–44 are averaged in float64 before applying threshold 0.5 (positive when the mean probability is at least 0.5). All 18,220 original test variants retain their frozen VariationID, label and protein mapping. The counts and feature profiles correspond to these ensembles in main Figure 4.

Corrected-sample group counts for raw ESM to full FPP:

Group	n	Positive rate
both correct	12,520	0.3460
both wrong	3,747	0.2906
improved/corrected	960	0.3812
degraded	993	0.0352
false negatives rescued	366	1.0000
false positives suppressed	594	0.0000
new false positives (TN to FP)	958	0.0000
new false negatives (TP to FN)	35	1.0000
persistent false negatives	1,089	1.0000
persistent false positives	2,658	0.0000

Corrected-sample group counts for architecture-matched CADD+AlphaMissense to full-FPP fusion:

Group	n	Positive rate
both correct	16,055	0.3117
both wrong	1,262	0.4136
improved/corrected	679	0.2504
degraded	224	0.5580
false negatives rescued	170	1.0000
false positives suppressed	509	0.0000
new false positives (TN to FP)	99	0.0000
new false negatives (TP to FN)	125	1.0000
persistent false negatives	522	1.0000
persistent false positives	740	0.0000

At threshold 0.5, the raw-ESM/full-FPP pair corrects 960 variants and degrades 993; degradation comprises 958 new false positives and 35 new false negatives. The fusion pair corrects 679 and degrades 224, including 99 new false positives and 125 new false negatives. These threshold-dependent counts are descriptive, and the sigmoid outputs are uncalibrated (Table 10).

In main Figure 4C–D, within-row centered and range-scaled means indicate which corrected group has the higher feature mean. Panel E uses uniform dot size and opacity; colors indicate only the direction of the FP-suppressed versus FN-rescued mean difference, without significance encoding. Panel F shows descriptive Hedges' g for rescued versus persistent false negatives. These features use different units, so raw differences are not comparable across features.

Selected false-positive-suppressed versus false-negative-rescued topology contrasts for raw ESM to full FPP:

Feature	FP mean	FN mean	Diff
degree bucket	0.7845	1.8443	-1.0598
k-core bucket	0.7525	1.5574	-0.8049
neighbour risk bucket	0.6431	1.0601	-0.4170
neighbour pathogenic rate	0.2891	0.3868	-0.0978
betweenness bucket	0.8771	1.5492	-0.6721
clustering coefficient	0.2595	0.1711	+0.0884
boundary role bucket	1.0859	1.3169	-0.2311

Selected false-positive-suppressed versus false-negative-rescued topology contrasts for architecture-matched CADD+AlphaMissense to full-FPP fusion:

Feature	FP mean	FN mean	Diff
degree bucket	1.5737	1.8059	-0.2322
k-core bucket	1.5639	1.6176	-0.0538
neighbour risk bucket	0.8035	1.7765	-0.9729
neighbour pathogenic rate	0.3344	0.4971	-0.1627
betweenness bucket	1.5639	1.6941	-0.1303
clustering coefficient	0.1818	0.2275	-0.0456

Feature	FP mean	FN mean	Diff
boundary role bucket	1.0314	0.7176	+0.3138

Baseline-error-stratified rescued false-negative contrasts for raw ESM:

Feature	Corrected subset mean	Persistent error mean	Hedges' g
Degree bucket	1.8443	1.5574	+0.5379
K-core bucket	1.5574	1.5390	+0.0329
Neighbour risk bucket	1.0601	1.0790	-0.0234
Neighbour pathogenic rate	0.3868	0.3778	+0.0735
Betweenness bucket	1.5492	1.5326	+0.0269
Clustering coefficient	0.1711	0.2338	-0.7111
Boundary role bucket	1.3169	1.0156	+0.3415
Within-community neighbour fraction	0.7823	0.7129	+0.3600

Baseline errors are stratified into false negatives and false positives, then corrected and persistent errors are contrasted within each stratum. Thus class label and baseline binary prediction agree within a contrast; continuous baseline scores are not matched and no caliper, nearest-neighbour or propensity matching is performed. The displayed FN contrasts contain 366 rescued and 1,089 persistent false negatives, except within-community neighbour fraction, for which 2 persistent-error values are missing (1,087 available). Missing values are excluded separately for each feature. Hedges' g uses the pooled within-group sample standard deviation with correction $1-3/[4(n1+n2)-9]$. Group means and effect sizes are variant-weighted descriptive summaries. Bucket encodings, including -1 for insufficient or unassigned context, are retained numerically; a bucket mean is not a biological effect magnitude.

Complete counts, positive/negative counts and protein counts for both pairs are supplied in `table6_all_transition_counts.csv`. All 11-feature FP-versus-FN contrasts and all 44 baseline-error-stratified contrasts (two model pairs, two error types and 11 features) are supplied in `table6_all_22_FP_FN_contrasts.csv` and `table6_all_44_error_stratified_contrasts.csv`, including actual feature-specific sample sizes and missing-value counts. Multiple variants from the same protein share network features and are not independent. These descriptive associations make no inferential significance or causal claim and do not eliminate residual difficulty confounding.

Supplementary Table 7. Negative-control audit

The formal supplementary table reports only the negative-control result that was rerun and re-materialized with complete final-audit provenance. Older random-feature and label-shuffling summaries are not used as formal evidence in this submission.

Control	Split	n	Val A	Val AP	Test A	Test AP
Feat.-shuffled FPP	Orig.	1	0.7080	0.4241	0.5868	0.3707

Val A and Test A denote validation and test AUC, respectively. Val AP and Test AP denote validation and test AUPRC. This control is used only as supplementary negative-control evidence and is not part of the main comparative claims.

Supplementary Table 8. Experimental configurations and run provenance

Supplementary Table 8A. Matched comparisons used in the main figures

Analysis	Fits and classifier seeds	Maximum epochs	Configuration
Original-split representations	3 per arm; 42–44; 9 for permutations	50	Common propagation and training protocol
Partial ablation, .25/.50/.75	3 per strategy/fraction; 42–44	50	Nine fixed intervention graphs; 27 fits
Partial ablation, 0 and 1	3 per endpoint; 42–44	50	Six endpoint fits; shared across strategies
Original-split fusion	3 per arm; 42–44; 9 for permutations	50	Same hidden-layer architecture and preprocessing
Homology-aware representations	3 per arm; 42–44	50	Raw ESM and full FPP
Homology-aware fusion	3 per arm; 42–44	50	Scores, raw-ESM fusion and full-FPP fusion
Topology and error profiles	Frozen matched ensembles	NA	No additional fitting for Figures 3B–F and 4

All graph arms in Table 8A use the full propagation convention, with self-loops added before symmetrization. Partial ablation removes unique undirected train-test protein pairs and all corresponding edge entries. Each interior strategy/fraction uses one fixed graph and three classifier seeds; dispersion across fits does not measure variation across graph masks. Random removals use seeds 292, 542 and 792 at fractions 0.25, 0.50 and 0.75, respectively, so the random masks are not nested. Confidence-based removals use real STRING combined scores and retain the original pair ordering and score-tie convention. The full-graph and fully ablated endpoints use the same three fits across all strategies, rather than independent endpoint graphs.

Supplementary Table 8B. Archived reference configurations

Analysis	Runs and seeds	Maximum epochs	Interpretation
Original raw ESM and full FPP repeated	3 per arm; 42, 43, 44	50	Matched repeated positive comparison
Historical reference pair	One retained estimate per arm	50	Bootstrap conditional on fitted reference predictions
Direct-edge ablation, representation-only	5; 3042-3046	30	Different seeds/budget from full graph
Partial ablation, fractions .25/.50/.75	3 per strategy/fraction	30	Seeds 5042 + 1000*fraction + repeat index; +100000 high-score-first; +200000 low-score-first
Partial ablation, fraction 0	Same full-graph reference reused	50	Not three independent endpoint observations
Partial ablation, fraction 1	Same five direct-ablation runs reused	30	Not new strategy-specific observations
Topology permutation, representation-only	3 graph seeds 6042-6044 x 3 classifier seeds (7042-7044, 7142-7144, 7242-7244)	30	Nine runs; unequal budget from full graph
Original matched fusion arms	3 classifier seeds 42-44 per arm	50	Same hidden-layer architecture, training protocol and preprocessing
Original permuted fusion	3 graphs x seeds 42-44	50	9 runs; bootstrap conditions on sampled graphs
Homology CADD+AM and raw-ESM fusion	3 classifier seeds 42-44 per arm	50	Repeated baseline means
Homology graph fusion	1 retained run	50	Descriptive only; asymmetric comparison

Table 8B documents the archived reference analyses in Tables 2–4 and 9. Their 30-epoch partial-ablation series and single-fit homology representation estimates are not the source of main Figure 1E or Figure 2D, which use the matched configurations in Table 8A and the results in Table 11. Figure 3B–F uses the matched original-split raw/full ensembles with fixed topology groups and protein-clustered intervals (Table 5). Figure 4 uses the matched original-split raw/full and score/full-fusion ensembles for its error profiles.

Supplementary Table 9. Historical post-hoc complete-case evaluation of matched fusion predictions

Both CADD and AlphaMissense scores were observed for 17,581 of 18,220 original test variants (639 excluded; pathogenic prevalence 32.0061%). Membership was matched by unique VariationID and verified against labels, genomic coordinates and reference/alternate alleles. Each frozen prediction vector reproduced its saved full-test AUC and average precision within 1e-6 before restriction. Models were not retrained; this tests sensitivity to test-set missing-score membership, not a complete-case training alternative.

Model	Runs	All-test AUC	Complete-case AUC	All-test AP	Complete-case AP
CADD+AM matched	3	0.9572	0.9597	0.9068	0.9118
+ raw ESM	3	0.9680	0.9694	0.9317	0.9344
+ edge-ablated FPP	3	0.9708	0.9720	0.9395	0.9420
+ permuted FPP	9	0.9694	0.9708	0.9355	0.9382
+ real FPP	3	0.9718	0.9730	0.9415	0.9438

Entries are across-run means. Per-run values and standard deviations are in revision_20261008/complete_case_matched_per_run.csv and complete_case_matched_summary.csv. AP is

non-interpolated average precision, the implementation used for the AUPRC-labelled metrics throughout this study. Ordering is preserved in this restricted subset; no independent generalization or equivalence claim follows.

Supplementary Table 10. Post-hoc threshold sensitivity of main Figure 4 matched ensembles

The same threshold is applied to both frozen ensembles at each of nine thresholds (0.1 to 0.9), without selecting a best threshold or refitting. VariationID, label and STRING protein ID are aligned to the same original test rows. The 0.5 counts reproduce main Figure 4A–B. Net corrections equal corrected minus degraded variants. These are descriptive test-set checks of uncalibrated class-weighted sigmoid scores, not clinical operating-point validation.

Threshold	ESM-FPP FN rescued	ESM-FPP FP suppressed	ESM-FPP net	Fusion FN rescued	Fusion FP suppressed	Fusion net
0.1	61	864	-701	27	1307	950
0.2	179	835	-555	73	981	747
0.3	245	562	68	84	799	610
0.4	166	476	122	114	648	491
0.5	366	594	-33	170	509	455
0.6	1201	414	264	273	435	501
0.7	598	207	-446	395	353	463
0.8	450	121	173	615	223	516
0.9	1178	3	950	912	125	654

Net corrections and the balance between false-negative rescue and false-positive suppression depend on the threshold. The detailed error-type interpretation at 0.5 therefore should not be generalized to all operating points. No calibration experiment has been performed. Complete partitions and all error-transition counts at all nine thresholds are supplied in `table10_all_threshold_transition_counts.csv`.

Remaining validation boundaries

Matched-seed/budget/operator representation controls and repeated homology graph fusion are completed (Tables 11–12). External-source evaluation and a second, physical-category interaction network are now evaluated in Methods 6 and Tables 14–17. These do not establish independent patients, prospective or predictor-development-disjoint evaluation, or full homology separation of the external source. Trained graph models, independent confirmation of selected topology subgroups and clinical calibration remain outside the current evidence. Binary labels discard graded function; clinical or functional confirmation remains necessary.

Source-data archive status

The compact source-data archive contains matched source data for main Figure 1B–F and Figure 2B–E, matched-ensemble topology metrics and protein-clustered intervals for Figure 3B–F, ESM-similarity source data for Figure 3A, matched-ensemble transition and feature data for Figure 4, and external-source and network-comparison data for Figure 5. The complete 177,213-row sample/split/cluster manifest, frozen node order, and core per-seed and external predictions are supplied in `canonical_analysis_20261025/public/`. The partial-ablation and homology-representation runs, endpoint-replication checks, canonical per-seed predictions, summary metrics and protein-bootstrap results are supplied in `remaining_unification_20261108/`. Tables 5, 6 and 10 use the original matched ensembles; their prediction sets are unchanged by the additional representation comparisons. Archived reference analyses retain their original source files and conventions. Figure-generation scripts, executed protocols, source manifests and checksums accompany the compact outputs. Large tensors and checkpoints are retained separately; third-party resources, including provider-controlled CADD and AlphaMissense score files, should be obtained from their original providers.

Supplementary Methods 5. Matched validation and restored runtime

The archived historical `conda_environment.yml` and `pip_freeze.txt` were retained unchanged with SHA-256 hashes. A separate runtime restored Python 3.10.8, PyTorch 2.1.2+cu118, torchvision 0.16.2+cu118, NumPy 1.26.4, pandas 2.3.3, scikit-learn 1.7.2 and SciPy 1.15.3. All 122 pip pins and 16 portable Python-library pins from the conda section were checked. Operating-system libraries and conda management packages were not a bitwise copy of the old base environment; the full resolved export is included. CUDA 11.8 is the PyTorch build, not a recovered historical driver/toolkit record. New runs used one NVIDIA RTX 4090.

The 18 staged input files were hash-verified against the local snapshot. Original model-input labels and protein indices matched every CSV row. Original and homology partitions covered the same VariationIDs. Both splits were protein-disjoint; saved homology cluster IDs were disjoint. Saved full FPP was numerically checked against recomputation in 16 feature columns (maximum absolute difference 1.49e-7). At the initial restoration check, six raw-ESM/full-FPP positive-control fits reproduced the historical variant-batched validation/test metrics and selected epochs to exported precision. This supported runtime recovery for those runs; current reporting subsequently applies the common node-lookup and aggregation convention described below. Ablated and all three permuted FPP tensors were regenerated using the same full operator, with graph permutations verified against seeds 6042–6044. ESM extraction and MMseqs clustering were not rerun.

Matched downstream classifiers use seeds 42–44, batch size 1,024, Adam learning rate 0.001 and weight decay 0.00001, class-weighted BCE, 512/128 hidden layers, dropout 0.3, maximum 50 epochs and patience five on strictly improving validation AUROC. Imputation uses training medians with an AlphaMissense missingness indicator; no z-score scaling was applied. Every fusion concatenates external scores before the protein representation, consistently across arms. This common ordering and isolated runtime mean the new fits should not be called bitwise replicas of every historical fit. GPU-resident node lookup changes storage, not feature values or row identity; test labels do not enter optimization or model selection. Test predictions are saved only after selecting each classifier by validation AUC.

The original-split representation comparison uses raw ESM, full FPP, ablated FPP (three fits each) and three topology permutations by three seeds (nine fits). Original-split fusion uses score-only, raw, full and ablated arms (three fits each) plus permuted fusion (nine). Homology fusion uses score-only, raw and full FPP (three fits each). Score-only and fusion models use the same hidden-layer architecture, but score-only models have fewer input-layer parameters. The smoke checks used four training batches plus validation, without test evaluation.

The matched STRING experiments comprise 81 distinct condition–seed combinations: the 48 core comparisons above, 27 partial-ablation interior fits and six homology-aware representation fits. Six additional full-graph/fully ablated endpoint fits bring the total to 87 fitting executions. The three frozen-full positive controls reproduced their saved metrics and selected epochs. Sparse GPU propagation was not bitwise reproducible across repeated evaluations. We retained the first regenerated fully ablated tensor and its first three fits, froze their checksums and checked the propagation independently in float64 on the CPU in 16 columns. These fully ablated fits differ numerically from the archived fits and supply both Figure 1D and

the shared fraction-1 endpoint in Figure 1E. Diagnostic regenerations were excluded from training. The full-graph endpoint is likewise shared across strategies. The additional fits used the same Python and numerical-library versions in a separate runtime: 122 exported pip-package pins and 16 portable conda-library pins were verified, together with pip, setuptools and wheel, for 141 verified package versions. Input checksums, frozen node order, all sample labels and split memberships, homology-cluster disjointness, and the full-graph propagation calculation were verified before fitting. No ESM extraction or homology clustering was repeated. Partial ablation retained 621,211 eligible undirected train-test pairs (1,242,422 edge entries) with complete STRING score coverage; each selected pair was removed in both directions, followed by recomputation of normalized propagation. The 25%, 50% and 75% conditions removed 155,303, 310,606 and 465,908 pairs, respectively. Graph masks were fixed independently of test labels, and every classifier was selected only by validation AUC.

Unified reporting. Tables 11–17 and main Figures 1B–F, 2B–E, 3B–F, 4 and 5 use the same inference and aggregation convention. For protein-only representation models, CPU evaluation in float32 with fixed batches of 1,024 nodes in the frozen node order produces one probability per protein, which is mapped to its variant rows. Fusion inputs remain variant-specific. Seed probabilities are converted to float64 before averaging, and AUPRC is calculated as non-interpolated average precision. Per-run means and sample standard deviations summarize individual classifier metrics; ensemble differences are calculated after averaging probabilities. The original reporting audit covered 21 representation checkpoints: on the original validation/test rows, the largest change from variant-batched probabilities was 3.58e-7 and the maximum within-protein spread decreased from 2.38e-7 to exactly zero. The 39 additional fitting executions use the same node-wise inference convention, with every mapped validation/test probability checked against the saved variant-batched output within 2e-6. The homology representation interval uses 2,000 paired protein-clustered resamples of the full-FPP and raw-ESM ensembles (Table 12), conditional on the fitted models and fixed graph. These intervals omit retraining and graph-sampling uncertainty and are not adjusted for multiple comparisons.

Supplementary Table 11. Current matched fitted-run summaries

Values use canonical per-run metrics and means ± sample SD (ddof=1) across classifier fits. Permutation rows are shown separately by graph; their three seeds are classifier repeats conditional on that graph. Tables 2–3, 8B and 9 retain separate reference sources. Figures 3B–F and 4 and Tables 5, 6 and 10 use the matched original-split ensembles described in Methods 5.

Supplementary Table 11A. Representation and fusion comparisons

Split	Arm	Fits	Test AUC	Test AUPRC
Homol. repr.	Raw ESM	3	0.7288 ± 0.0039	0.2464 ± 0.0040
Homol. repr.	Full FPP	3	0.7571 ± 0.0022	0.2751 ± 0.0022
Homology	Scores (512/128)	3	0.9755 ± 0.0001	0.8403 ± 0.0006
Homology	+ full FPP	3	0.9752 ± 0.0007	0.8369 ± 0.0060
Homology	+ raw ESM	3	0.9727 ± 0.0003	0.8253 ± 0.0016
Original	Ablated FPP	3	0.7694 ± 0.0031	0.6035 ± 0.0056
Original	Full FPP	3	0.7950 ± 0.0016	0.6485 ± 0.0117
Original	Permuted FPP (graph 0)	3	0.7807 ± 0.0046	0.6096 ± 0.0019
Original	Permuted FPP (graph 1)	3	0.7751 ± 0.0148	0.6014 ± 0.0049
Original	Permuted FPP (graph 2)	3	0.7851 ± 0.0049	0.6199 ± 0.0013
Original	Raw ESM	3	0.7835 ± 0.0063	0.6117 ± 0.0066
Orig. fusion	Scores (512/128)	3	0.9572 ± 0.0001	0.9072 ± 0.0002
Orig. fusion	+ ablated FPP	3	0.9704 ± 0.0004	0.9394 ± 0.0009
Orig. fusion	+ full FPP	3	0.9716 ± 0.0007	0.9409 ± 0.0017
Orig. fusion	+ permuted (graph 0)	3	0.9699 ± 0.0002	0.9351 ± 0.0014
Orig. fusion	+ permuted (graph 1)	3	0.9690 ± 0.0005	0.9354 ± 0.0009
Orig. fusion	+ permuted (graph 2)	3	0.9688 ± 0.0003	0.9357 ± 0.0005
Orig. fusion	+ raw ESM	3	0.9682 ± 0.0011	0.9326 ± 0.0037

Core per-run metrics, seeds and selected epochs are recorded in `canonical_analysis_20261025/per_run_metrics.csv`, with training histories in `matched_validation_20261024/`. Per-run metrics for the homology representations, partial-ablation curve and updated fully ablated representation endpoint are supplied in `remaining_unification_20261108/`. Pooled permutation SD mixes graph and classifier dispersion and is descriptive. Paired seed differences are retained with their corresponding prediction sets.

Supplementary Table 11B. Matched partial train-test pair ablation

Removal strategy	Fraction	Fits	Test AUC	Test AUPRC
Random	0.00	3	0.7950 ± 0.0016	0.6485 ± 0.0117
Random	0.25	3	0.7929 ± 0.0108	0.6448 ± 0.0121
Random	0.50	3	0.7857 ± 0.0089	0.6362 ± 0.0134
Random	0.75	3	0.7858 ± 0.0070	0.6350 ± 0.0039
Random	1.00	3	0.7694 ± 0.0031	0.6035 ± 0.0056
High-score first	0.00	3	0.7950 ± 0.0016	0.6485 ± 0.0117
High-score first	0.25	3	0.7892 ± 0.0010	0.6397 ± 0.0087
High-score first	0.50	3	0.7844 ± 0.0007	0.6323 ± 0.0046
High-score first	0.75	3	0.7774 ± 0.0057	0.6156 ± 0.0085
High-score first	1.00	3	0.7694 ± 0.0031	0.6035 ± 0.0056
Low-score first	0.00	3	0.7950 ± 0.0016	0.6485 ± 0.0117
Low-score first	0.25	3	0.7897 ± 0.0042	0.6389 ± 0.0041
Low-score first	0.50	3	0.7883 ± 0.0084	0.6350 ± 0.0030
Low-score first	0.75	3	0.7900 ± 0.0153	0.6438 ± 0.0209
Low-score first	1.00	3	0.7694 ± 0.0031	0.6035 ± 0.0056

Entries are means ± sample SD across classifier seeds 42–44. Each interior condition uses one fixed graph; SD therefore describes classifier variation conditional on that graph. The fraction-0 and fraction-1 rows share the same full-graph and fully ablated fits across strategies. The 15 displayed points correspond to 11 distinct graph conditions and 33 classifier fits, including six endpoint fits.

Supplementary Table 12. Current protein-clustered paired ensemble margins

Probabilities are converted to float64 and averaged within arm before computing metrics; the permutation ensemble averages all three saved graphs and three classifier seeds. Each of 2,000 replicates samples test proteins with replacement and retains all variants with multiplicity. The random-number generator restarts at seed 20261024 for each population, and all arm comparisons within that population use the same sampled protein multiplicities. Intervals condition on fitted models and sampled permutations; they omit retraining, new graphs, cluster reassignments and external-cohort uncertainty. Comparisons are exploratory, without multiplicity adjustment. A zero-crossing interval does not establish equivalence.

Split	Comparison	ΔAUC (95% CI)	ΔAUPRC (95% CI)
Homol. repr.	full FPP − raw ESM	+0.0288 (+0.0185, +0.0403)	+0.0304 (+0.0131, +0.0554)
Original	full FPP − raw ESM	+0.0107 (-0.0218, +0.0426)	+0.0412 (-0.0106, +0.0767)
Original	full FPP − ablated FPP	+0.0264 (+0.0078, +0.0521)	+0.0565 (+0.0196, +0.0778)
Original	full FPP − permuted FPP	+0.0118 (-0.0229, +0.0462)	+0.0430 (-0.0229, +0.0883)
Orig. fusion	full FPP − scores	+0.0147 (+0.0065, +0.0226)	+0.0344 (+0.0156, +0.0558)
Orig. fusion	full FPP − raw ESM	+0.0032 (+0.0008, +0.0062)	+0.0076 (+0.0030, +0.0141)
Orig. fusion	full FPP − ablated FPP	+0.0013 (-0.0004, +0.0032)	+0.0014 (-0.0015, +0.0047)
Orig. fusion	full FPP − permuted FPP	+0.0020 (+0.0001, +0.0045)	+0.0050 (+0.0002, +0.0110)

Split	Comparison	ΔAUC (95% CI)	ΔAUPRC (95% CI)
Homol. fusion	full FPP − scores	+0.0001 (-0.0019, +0.0020)	-0.0012 (-0.0205, +0.0157)
Homol. fusion	raw ESM − scores	-0.0025 (-0.0048, -0.0004)	-0.0135 (-0.0343, +0.0045)
Homol. fusion	full FPP − raw ESM	+0.0027 (+0.0015, +0.0040)	+0.0123 (+0.0055, +0.0194)

Variant-level resampling of the original full-FPP versus raw-ESM ensembles is also supplied, using 2,000 replicates. It ignores within-protein dependence and is used only to contrast uncertainty estimates. The weighted metric implementation was checked against scikit-learn on the actual predictions, including random protein multiplicities.

Supplementary Table 13. Current complete-case test-subset sensitivity

Both external scores are observed for 17,581 original test variants. Canonical current predictions are evaluated on the full test set and then restricted without retraining. This addresses test-membership sensitivity, not complete-case training or independent validation. Entries are means of per-run metrics; the permutation row pools three graphs by three classifier seeds (n=9).

Arm	Fits	All-test AUC	Complete AUC	All-test AUPRC	Complete AUPRC
Scores (512/128)	3	0.9572	0.9597	0.9072	0.9122
+ raw ESM	3	0.9682	0.9696	0.9326	0.9353
+ ablated FPP	3	0.9704	0.9714	0.9394	0.9414
+ permuted FPP	9	0.9692	0.9706	0.9354	0.9381
+ full FPP	3	0.9716	0.9726	0.9409	0.9430

Supplementary Methods 6. External-source and second-network validation

Frozen design and provenance. The extension protocol was fixed before new fitting or external scoring. It reused the verified original and homology partitions, actual 19,622-node order and 1,280-dimensional ESM features; neither embedding extraction nor homology clustering was repeated. The external source was UniProtKB/Swiss-Prot HUMSAVAR release 2026_03 (2 September 2026), with corresponding canonical human reviewed sequences. HUMSAVAR is separately curated, not an independent patient cohort; it can share literature or annotations with ClinVar. The frozen benchmark's unresolved content-release date also precludes a temporal-independence claim.

External-source construction. LP/P and LB/B records defined positive and negative labels; uncertain significance was excluded. Each retained standard single amino-acid substitution required a correct reference residue and exact whole-canonical-sequence identity to a uniquely mapped frozen STRING node. Duplicate variants were collapsed and conflicting labels excluded. The union of accession-plus-substitution, STRING-node-plus-substitution, gene-plus-substitution and dbSNP identifiers excluded overlaps with all 177,213 original variants, including the test set. Before inference, mapping review identified seven CALM1/CALM3 records on one STRING node associated with multiple retained accessions; all seven were excluded, without arbitrary accession selection. Primary and strict subset membership was unchanged. All final indices were checked against the actual model-input protein-to-index mapping; no replacements were needed.

The source contained 85,015 rows. Sequential exclusion counts were: uncertain significance, 11,718; no full-sequence identity, 11,386; original-benchmark overlap, 19,771; redundant duplicate, 299; absent accession alias, 633; conflicting duplicate, 35; ambiguous identical graph nodes, 265; nonstandard substitution, one; and cross-accession node collision, seven. This left 40,900 variant-disjoint records. The primary cohort excluded original training/validation labelled proteins and their exact sequences; strict additionally excluded original test labelled proteins and their exact sequences (Table 14). Strict is nested within primary, not a second independent replication. The all-record population includes training proteins and is descriptive only. Whole-sequence exclusions do not separate every distant homolog. All evaluated proteins remain nodes in the frozen transductive feature graph.

Alternative interaction network. BioGRID 5.0.262 supplied the second network (October 2026 release; archive compiled 25 September 2026). We retained human–human physical-category records, excluding Affinity Capture-RNA and Protein-RNA methods. Entrez identifiers had to map uniquely to a frozen STRING

node; eight ambiguous mapped identifiers were excluded. Self interactions were removed, unordered pairs deduplicated, and both edge directions retained. The graph has 931,458 undirected edges on all 19,622 nodes, including 1,402 isolates; no test rows were removed for absent BioGRID edges. Of its pairs, 526,635 (56.54%) also occur in STRING (Table 15). Physical-category records include complex association or proximity, not only direct binary binding. STRING incorporates BioGRID evidence, so this is alternative-network robustness rather than evidence-independent replication. Coverage, density, release and evidence composition differ together.

Both networks used the common Methods 2 operator: add loops before symmetrization, retain summed edge multiplicities, apply weighted symmetric normalization, and average raw, one-hop and two-hop features. BioGRID checks confirmed bounded actual tensor indices, bidirectionality, no duplicate directed edges and no pre-existing self edges. Its propagated matrix was finite with shape 19,622 × 1,280. Isolates differed from raw features by at most 9.54e-7; independent 16-column propagation agreed within 1.20e-7. No labels were used in propagation.

Fitting and inference. Nine new BioGRID classifiers were fitted: original-split representation, original-split score fusion and homology-split score fusion, each with seeds 42–44. They used the same 512/128 MLP, dropout 0.3, weighted BCE, Adam learning rate 0.001, weight decay 0.00001, batch size 1,024, maximum 50 epochs and patience five as Methods 5. Strict validation AUC improvement selected the checkpoint; test and external outcomes did not select hyperparameters, epochs or membership. Existing raw, STRING and score baselines were reused without retuning. All original and homology test rows were retained.

Six retained raw/STRING checkpoints reproduced recorded validation/test metrics before external evaluation. External inference applied original-split raw, STRING and BioGRID representation classifiers to all three populations, yielding 27 population/model/seed combinations. No external classifier was fitted. External CADD/AlphaMissense fusion was not evaluated because genuine matched clinical scores were unavailable. Each protein-only model assigns the same score to all variants of a protein: this tests protein-context discrimination across the cohort, not discrimination among substitutions within a protein. Protein-defined HUMSAVAR substitutions need not all be genomic single-nucleotide variants.

Estimation and verification. Table 16 reports mean and sample SD (ddof=1) across all three seeds; Table 17 uses metrics of mean predicted probabilities across seeds. Paired differences use 2,000 protein-cluster bootstrap replicates, restarting seed 20261024 separately for each population. Each replicate samples proteins with replacement, retaining all variants with multiplicity; the same samples pair arms within each population. Percentile 95% intervals condition on fitted ensembles and frozen graphs, omitting retraining, graph-source variation and new cohorts. Comparisons are exploratory without multiplicity adjustment. Zero-crossing intervals do not establish equivalence.

The same float64 averaging and non-interpolated average-precision calculation are used for Tables 12 and 17. Canonical node lookup also gives exact same-protein scores for all 27 external representation prediction combinations. These were recomputed from the same frozen checkpoints; the largest probability difference from the earlier external node-batched scores was 4.47e-7. The external-validation reporting audit covers 57 frozen checkpoints, 90 per-seed metric rows, 30 ensemble summaries and 27 paired protein-bootstrap comparisons, with no new fitting. All 2,000 replicates were finite for every comparison. An independent scikit-learn calculation reproduced all 2,000 replicates for five key comparisons: original full FPP versus raw ESM, original fusion full FPP versus raw fusion, homology full FPP versus scores, and primary and strict external STRING versus raw ESM. All corresponding interval endpoints agreed within 2.17e-16; 14 synthetic tied-score/weight cases also passed. Other paired comparisons were checked through their point estimates, finite counts and shared metric implementation. Current compact outputs are in `canonical_analysis_20261025/`; earlier analyses remain dated archives. Tensors, checkpoints and large provider resources are retained separately.

Supplementary Table 14. External-source populations after overlap exclusions

Population	Variants	Proteins	Positive	Negative	Additional protein exclusion
Primary	10,785	2,498	4,301	6,484	Original train/validation and identical sequences
Strict, nested	6,934	1,713	2,555	4,379	All original labelled proteins and identical sequences
All variant-disjoint	40,900	9,840	15,950	24,950	None; descriptive only

All rows passed the same variant-overlap, sequence and reference-residue checks. Counts are unique STRING proteins; a protein can contain both positive and negative variants. Primary includes 3,851 variants on 785 original test proteins and 6,934 variants on 1,713 previously unlabelled graph proteins. Excluding training proteins from evaluation does not remove them from graph propagation.

Supplementary Table 15. Alternative network on the frozen node universe

Quantity	Value
Frozen nodes, both networks	19,622
Retained STRING undirected pairs	6,857,702
BioGRID undirected pairs	931,458
BioGRID directed entries before loops	1,862,916
BioGRID isolates retained	1,402
Shared undirected pairs	526,635
BioGRID-only pairs	404,823
BioGRID pairs in STRING	56.54%
Pair-set Jaccard index	0.07251
Original test variants with a BioGRID edge	18,006 / 18,220
Homology test variants with a BioGRID edge	17,748 / 18,220

From 2,950,299 BioGRID source rows, 1,319,164 were human–human, 1,298,722 were physical-category, 1,262,564 remained after RNA-method exclusions, and 1,241,565 had both endpoints uniquely mapped. Removing 9,103 mapped self-interaction records left 1,232,462 evidence rows; 301,004 duplicate pair records were collapsed. Propagation graphs differ from the thresholded topology-description graph in Table 1.

Supplementary Table 16. Three-seed performance in the extension

Entries are mean ± sample SD across seeds 42–44, not metrics of ensemble-averaged predictions. AUPRC denotes non-interpolated average precision. All arms have three classifier seeds; external models are frozen original-split checkpoints.

Supplementary Table 16A. External-source representation performance

Population	Representation	AUC	AUPRC
Primary	Raw ESM	0.6932 ± 0.0067	0.5719 ± 0.0088
Primary	STRING FPP	0.7304 ± 0.0042	0.6146 ± 0.0074
Primary	BioGRID FPP	0.7025 ± 0.0116	0.5816 ± 0.0064
Strict	Raw ESM	0.7267 ± 0.0060	0.5696 ± 0.0119
Strict	STRING FPP	0.7616 ± 0.0048	0.6213 ± 0.0117
Strict	BioGRID FPP	0.7292 ± 0.0033	0.5840 ± 0.0064
All (descriptive)	Raw ESM	0.7674 ± 0.0027	0.6588 ± 0.0051
All (descriptive)	STRING FPP	0.8007 ± 0.0091	0.6944 ± 0.0130
All (descriptive)	BioGRID FPP	0.7888 ± 0.0229	0.6860 ± 0.0307

Primary and strict STRING-versus-raw ensemble margins are positive (Table 17A); BioGRID-versus-raw intervals cross zero in both. All-record results include training proteins and cannot substitute for the primary evaluation.

Supplementary Table 16B. Frozen-benchmark network comparison

Setting	Arm	AUC	AUPRC
Representation	Raw ESM	0.7835 ± 0.0063	0.6117 ± 0.0066
Representation	STRING FPP	0.7950 ± 0.0016	0.6485 ± 0.0117
Representation	BioGRID FPP	0.7592 ± 0.0133	0.6043 ± 0.0083
Original fusion	Scores	0.9572 ± 0.0001	0.9072 ± 0.0002
Original fusion	Scores + raw	0.9682 ± 0.0011	0.9326 ± 0.0037
Original fusion	Scores + STRING	0.9716 ± 0.0007	0.9409 ± 0.0017
Original fusion	Scores + BioGRID	0.9693 ± 0.0013	0.9344 ± 0.0038
Homology fusion	Scores	0.9755 ± 0.0001	0.8403 ± 0.0006
Homology fusion	Scores + raw	0.9727 ± 0.0003	0.8253 ± 0.0016
Homology fusion	Scores + STRING	0.9752 ± 0.0007	0.8369 ± 0.0060
Homology fusion	Scores + BioGRID	0.9730 ± 0.0002	0.8265 ± 0.0013

Scores denotes architecture-matched CADD + AlphaMissense. Only BioGRID arms were newly fitted in this extension; comparator seed metrics reproduce the corresponding current matched rows in Table 11.

Supplementary Table 17. Paired ensemble margins with protein-clustered 95% intervals

All 2,000 replicates were finite in every comparison. These are differences between metrics of averaged predictions, not differences between the seed means in Table 16. Five decimals preserve the homology-versus-scores upper AUC bound near zero. No multiple-comparison correction was applied.

Supplementary Table 17A. External-source representation margins

Population	Comparison	ΔAUC (95% CI)	ΔAUPRC (95% CI)
Primary	STRING − raw	+0.03748 (+0.01733, +0.05638)	+0.04632 (+0.01514, +0.07446)
Primary	BioGRID − raw	+0.01248 (-0.01319, +0.03504)	+0.01506 (-0.01183, +0.04465)
Primary	BioGRID − STRING	-0.02500 (-0.05134, -0.00154)	-0.03126 (-0.06281, +0.00399)
Strict	STRING − raw	+0.03518 (+0.00339, +0.06667)	+0.05496 (+0.01109, +0.09620)
Strict	BioGRID − raw	+0.00940 (-0.02499, +0.04353)	+0.02269 (-0.01645, +0.06507)
Strict	BioGRID − STRING	-0.02578 (-0.05818, +0.00573)	-0.03227 (-0.07441, +0.01829)
All (descriptive)	STRING − raw	+0.03482 (+0.02766, +0.04172)	+0.03996 (+0.02879, +0.05083)
All (descriptive)	BioGRID − raw	+0.02957 (+0.02161, +0.03721)	+0.03850 (+0.02859, +0.04874)
All (descriptive)	BioGRID − STRING	-0.00525 (-0.01275, +0.00226)	-0.00146 (-0.01218, +0.01039)

Supplementary Table 17B. Original and homology benchmark margins

Setting	Comparison	ΔAUC (95% CI)	ΔAUPRC (95% CI)
Representation	BioGRID − raw	-0.02012 (-0.04803, +0.01005)	-0.00249 (-0.05396, +0.03630)
Representation	BioGRID − STRING	-0.03079 (-0.06916, +0.00407)	-0.04371 (-0.08938, -0.00108)
Original fusion	BioGRID − scores	+0.01242 (+0.00341, +0.02100)	+0.02788 (+0.00701, +0.05019)
Original fusion	BioGRID − raw	+0.00092 (-0.00115, +0.00364)	+0.00105 (-0.00371, +0.00691)
Original fusion	BioGRID − STRING	-0.00228 (-0.00420, -0.00058)	-0.00650 (-0.01164, -0.00233)
Homology fusion	BioGRID − scores	-0.00204 (-0.00421, -0.00004)	-0.01176 (-0.03147, +0.00596)
Homology fusion	BioGRID − raw	+0.00051 (-0.00077, +0.00169)	+0.00177 (-0.00629, +0.01049)
Homology fusion	BioGRID − STRING	-0.00216 (-0.00357, -0.00090)	-0.01052 (-0.01884, -0.00182)

Original-split BioGRID fusion improved over scores alone, but incremental margins over raw-ESM fusion crossed zero for both metrics. BioGRID fusion was below STRING fusion in both splits under these conditional intervals. The homology BioGRID-minus-scores AUC interval ends only 0.00004 below zero, whereas its AUPRC interval crosses zero. This exploratory result does not establish stable harm from physical interactions. The external-source and second-network comparisons support transfer of the particular STRING representation under the stated exclusions and show network dependence; they do not establish a universal benefit from graph augmentation.